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ORIGINAL ARTICLE

Efficacy and safety of ropeginterferon alfa-2b in Japanese patients with polycythemia vera: an open-label, single-arm, phase 2 study

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Abstract
Ropeginterferon alfa-2b is a novel, site-selective, monopegylated recombinant human interferon alfa-2b. Safety and efficacy of ropeginterferon alfa-2b for the treatment of polycythemia vera were demonstrated in clinical studies conducted in European countries, but clinical studies in Japanese patients are lacking. This phase 2, open-label, multicenter, single-arm study investigated the safety and efficacy of ropeginterferon alfa-2b in 29 Japanese patients with polycythemia vera including young patients and patients with low thrombosis risk who are difficult to receive guideline-based standard treatments. The primary outcome of durable complete hematologic response without phlebotomy at months 9 and 12 was achieved by 8/29 (27.6%) patients. The fastest complete hematologic response was observed at week 12. A corresponding reduction in the *JAK2* V617F allele burden from baseline to 52 weeks was also observed (mean ± standard deviation = − 19.2% ± 22.6%). No new safety concerns were identified in Japanese patients when compared with previous studies of ropeginterferon alfa-2b in European populations; the most common treatment-related adverse events were alopecia (55.2%), fatigue (27.6%) and influenza-like illness (27.6%). Most treatment-related adverse events were mild or moderate, with none of grade ≥ 3. Ropeginterferon alfa-2b is a safe and efficacious treatment option in Japanese patients with polycythemia vera.

Keywords Polycythemia vera · *JAK2* V617F allele burden · Ropeginterferon alfa-2b · Hematologic response · Molecular response

Introduction

Polycythemia vera (PV) is a *BCR-ABL*-negative myeloproliferative neoplasm characterized by the overproduction of mature erythrocytes associated with thrombohemorrhagic complications and harboring Janus kinase (*JAK*)*2* driver mutations in most cases [1, 2]. PV is typically diagnosed in individuals aged ≥ 65 years, but can also occur in younger patients (≤ 45 years) [3]. The long-term risks associated with PV include thromboembolic events, hemorrhage, and the propensity for a patient's disease to transform into myelofibrosis or secondary acute myeloid leukemia (sAML) [1, 4, 5]. The lifetime prognosis for patients with PV is significantly worse than for healthy individuals due to these associated risks [6–8]. The cumulative incidences of progression into myelofibrosis or sAML after diagnosis of PV are 2% and 1%, respectively, at 5 years; 9% and 2% at 10 years; and 19% and 7% at 15 years [9]. Furthermore, a high white blood cell (WBC) count or thrombosis at the time of initial diagnosis is particularly associated with a poor prognosis [10]. In younger patients with PV (≤ 45 years), rates of vascular complications have been found to be comparable with patients aged ≥ 65 years, despite lower WBC counts in the younger age group [11]. Moreover, the frequencies of transformation into myelofibrosis or sAML were also reported to be similar in both young and older patient groups [11].

Current treatment options for PV include phlebotomy and low-dose aspirin as first-line therapy to correct abnormal blood viscosity and reduce the risk of vascular events, regardless of the patient's thrombosis risk level. For high-risk patients, treatment recommendations also include cytoreductive therapy with either hydroxyurea (HU) or peginterferon alfa-2a, and second-line treatment with an alternate cytoreductive agent or the dual JAK1/JAK2

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inhibitor ruxolitinib, based on overseas guidelines [12–17]. However, there are no interferon preparations (including peginterferon alfa-2a) approved as an option for cytoreductive therapy for the treatment of PV in Japan [18], and the recommendation for cytoreductive therapy (i.e., HU) is limited to high-risk patients. Furthermore, the current standard cytoreductive treatment options such as HU are not appropriate for certain patient populations, including younger individuals and patients desiring to bear children [19]. In addition, there is controversy over whether HU may increase the risk of transformation to sAML [20–22]. Moreover, ruxolitinib as a second-line cytoreductive treatment for patients who cannot tolerate or are resistant to HU faces concerns such as the increased risk of infection and secondary malignancies [23], although another retrospective study failed to confirm this association [24].

A site-selective pegylated protein, ropeginterferon alfa-2b is a single, chemically homogenous isomer, and as such, is expected to have a better safety profile and improved pharmacokinetic properties compared with other pegylated interferon products that contain multiple isomers that may contribute to the development of adverse effects [25]. Pegylation extends the half-life of biologics by inhibiting their degradation in the body, allowing them to remain active for longer than non-pegylated products, and in turn, enabling longer dosing intervals (once every 2 weeks in the case of ropeginterferon alfa-2b) [26].

Several clinical studies, including PEGINVERA (NCT01193699) [27–29], PROUD-PV (NCT01949805) and PROUD-CONTINUATION-PV (NCT02218047) [30], and PEN-PV (NCT02523638) [31], have demonstrated ropeginterferon alfa-2b to be a safe and efficacious treatment option for patients with PV. In phase 3 clinical studies, ropeginterferon alfa-2b was effective in inducing hematologic responses and was well tolerated. In patients receiving ropeginterferon alfa-2b, responses to treatment continued to increase over time with the notable improvement compared with those of HU at 36 months [30, 31]. On the basis of these positive clinical data, ropeginterferon alfa-2b was approved in Europe, Taiwan, and Korea for the treatment of PV without symptomatic splenomegaly [32–34]. Most recently, ropeginterferon alfa-2b also received approval in the US for the treatment of polycythemia vera [35].

Ropeginterferon alfa-2b is currently not available in Japan. Although the pharmacokinetics (PK) of ropeginterferon alfa-2b has been evaluated in healthy Japanese volunteers [36], clinical studies in Japanese patients with PV have not previously been conducted. Thus, the aim of this phase 2, single-arm study was to investigate the efficacy and safety of ropeginterferon alfa-2b in Japanese patients with PV to assess the clinical utility of this cytoreductive therapy as a treatment option for Japanese patients.

Materials and methods

Patients

The general inclusion criteria were men or women aged $\geq$ 20 years at the time of informed consent; diagnosis of PV according to the World Health Organization 2008 [37] or 2016 [38] criteria; cumulative total HU treatment duration < 3 years at screening; adequate hepatic function (total bilirubin $\leq 1.5 \times$ upper limit of normal [ULN]); international normalized ratio (INR) $\leq 1.5 \times$ ULN; albumin > 3.5 g/dL; alanine aminotransferase $\leq 2.0 \times$ ULN; aspartate aminotransferase $\leq 2.0 \times$ ULN; hemoglobin $\geq$ 10 g/dL; neutrophil count $\geq 1.5 \times 10^9$/L; serum creatinine $\leq 1.5 \times$ ULN; Hospital Anxiety and Depression Scale (HADS) score 0–7 on both subscales; and use of birth control until 28 days following the last dose of the study drug. Patients for whom the current standard of treatment is difficult to apply were also included (i.e., younger patients for whom long-term treatment is anticipated; patients categorized as low risk, but for whom cytoreduction is recommended due to disease-related signs and symptoms; patients with HU intolerance [modified European LeukemiaNet criteria [39]]).

For cytoreduction-naïve patients only, an additional inclusion criterion was PV requiring cytoreductive treatment, defined as one or more of the following: $\geq$ 1 documented major cardiovascular PV-related event; poor tolerance of phlebotomy or frequent need of phlebotomy; platelet count > 1000×10^9/L or WBC count > 10×10^9/L at two measurements within the month prior to treatment start; and the manifestation of disease-related signs and symptoms (e.g., headache, dizziness, pruritus, night sweats, fatigue, erythromelalgia, vision disorders, scintillating scotoma, early satiety, or abdominal distension).

Key exclusion criteria were symptomatic splenomegaly; previous use of interferon alpha or contraindications/hypersensitivity to interferon alpha; documented history of resistance to HU; the presence of circulating blasts in the peripheral blood within the prior 3 months; and any serious clinical condition that may affect study participation or patient safety.

All patients provided written informed consent for study participation.

Study design and treatments

This was a phase 2, open-label, multicenter, single-arm study to investigate the efficacy and safety of ropeginterferon alfa-2b in Japanese patients with PV. The study rationale, design, and initial baseline data were previously published in a brief report [40]. The study was conducted between December 2019 and April 2021 at eight sites. The design of the study included a screening phase of 28 days, a 52-week treatment phase with visits every 2 weeks, and a safety follow-up for 28 days after the end of treatment. An extension study (NCT04655092) is currently ongoing.

Data for the efficacy evaluation were collected at baseline and at 12, 24, 36, and 52 weeks. Samples for exposure analysis were collected at week 0 and week 28 at 0 (pre-dose), 48, 96, and 168 h. Trough PK samples were collected at every visit.

Ropeginterferon alfa-2b was administered subcutaneously once every 2 weeks, starting at 100 µg. The dose was increased by 50 µg per visit until a response was achieved or to a maximum of 500 µg, and then remained fixed for the remainder of the study. Adjustments were allowed based on tolerability. For patients already receiving cytoreductive therapy, the starting dose was 50 µg, and the prior therapies were reduced as the dose of ropeginterferon alfa-2b increased. For doses ≥ 300 µg, patients were required to be hospitalized for monitoring for 2 days from the time of administration. However, after the safety of the 500-µg dose was confirmed in five of the six first evaluable patients by the coordinating principal investigator, the principal investigator at each study site, and the sponsor's medical monitor, the 2-day hospitalization period was deemed unnecessary, and patients could receive doses exceeding 300 µg without hospitalization for the remainder of the study.

All patients received low-dose aspirin (75–150 mg/day), or another prophylactic antithrombotic agent if aspirin was contraindicated. Prophylactic acetaminophen or another non-steroidal anti-inflammatory drug was recommended to avoid injection site reactions or flu-like symptoms. Patients experiencing mild or moderate thromboembolic events could receive anticoagulant treatment and severe events would trigger study discontinuation. Phlebotomy was performed when the hematocrit value was ≥ 45%; the recommended volume was 200–400 mL (or 100–200 mL for elderly patients or those with cardiovascular disorders).

The study was registered at ClinicalTrials.gov (NCT04182100). Ethical approval was granted by the institutional review boards at each site, and all procedures were conducted in accordance with the Declaration of Helsinki and all other applicable ethical and legal requirements.

Study objectives

The primary objective of this study was to demonstrate the safety and efficacy of ropeginterferon alfa-2b in terms of complete hematologic response (CHR) without phlebotomy in Japanese PV patients for whom the current standard therapy is difficult to apply. The phlebotomy-free responder rate was defined as the proportion of patients with CHR and no phlebotomies during the previous 12 weeks. CHR was defined as hematocrit < 45%, normalization of WBC count ($\leq 10 \times 10^9$/L), and normalization of platelet count ($\leq 400 \times 10^9$/L). To meet the primary endpoint, responders were required to have a CHR and no requirement for phlebotomy at weeks 36 and 52.

Secondary objectives were to demonstrate safety and efficacy in terms of changes in hematocrit, WBC count, platelet count, spleen size, molecular response based on the *JAK2* V617F allele burden at baseline, time to phlebotomy cessation, number of phlebotomy procedures required, time to response, duration of response maintenance, CHR rates according to prior HU treatment, proportion of patients without thrombotic events, and relationship between exposure and key efficacy and safety endpoints using an exposure–response analysis. Definitions of molecular response were based on the criteria published in 2009 [41] to compare results with those of the PROUD-PV study. An additional analysis based on updated criteria [42] was also conducted. Spleen index (cm^2) was defined as the maximum length on the longitudinal sonogram (cm) × width on the transverse sonogram (cm). Quantitative *JAK2* V617F allele burden measurements were conducted by LSI Medience (Tokyo, Japan), and determination of ropeginterferon alfa-2b concentration was conducted by PPC (Taipei, Taiwan) using a commercial enzyme-linked immunosorbent assay (ELISA) kit for human interferon alpha (VeriKine-HS Human Interferon Alpha All Subtype ELISA Kit, PBL Assay Science, Piscataway, NJ, USA).

Safety outcomes

Treatment-emergent adverse events (TEAEs) were categorized using the Medical Dictionary for Regulatory Activities v23.0 and assessed for severity using the Common Terminology Criteria for Adverse Events v5.0. Incidence, causality, and severity of TEAEs of special interest (e.g., cardiovascular, hemorrhagic, thrombotic, and psychiatric) were also noted. Additional safety parameters included assessment of vital signs, clinical laboratory tests, physical examinations, electrocardiography, echocardiography, chest X-ray, Eastern Cooperative Oncology Group (ECOG) performance status (PS), and ocular examination.

Statistical methods

Sample size calculations were based on the projected population of Japanese patients with PV for whom the current standard of treatment is difficult to apply, and the CHR rates determined in the PROUD-PV study at 3, 6, 9, and 12 months (11%, 27%, 36% and 43%, respectively) [30]. Assuming that the lower limit of the 95% confidence interval (CI) for the CHR rate was > 11% and that the slope of the linear regression model of the CHR time course was greater than 0 (positive trend), the power of the study at 9 months would be 85% with 25 patients. Allowing for the possibility of drop-outs, it was considered that 30 patients should be enrolled.

The intention-to-treat (ITT) and safety populations were identical, and included all patients who received at least one dose of the study drug. Categorical data were reported using numbers and percentages and continuous data using mean, standard deviation (SD), median, and range. For time-to-event data, Kaplan–Meier methodology was used to estimate the distribution and the median (with 95% CIs). For the estimation of drug exposure, PK measurements included the area under the curve from time zero to the last measurable concentration ($AUC_{0\text{-}t}$), maximum concentration (C_{max}), time to C_{max} (t_{max}), and half-life ($t_{1/2}$). Details regarding the power model regressions and equivalence criteria are described in Supplementary Data 1.

There was no imputation for missing data. All statistical calculations were performed using SAS v9.4 or higher (SAS Institute Inc., Cary, NC, USA).

Results

Patients

The disposition of patients is shown in Fig. 1. Of the 32 patients screened, three were not enrolled (two did not meet the inclusion/exclusion criteria and one withdrew consent). Thus, the ITT population and the safety population both comprised 29 patients. Of these, 27 patients completed the study; one patient withdrew consent, and one discontinued due to a TEAE (silent thyroiditis that did not recover within 8 weeks). All 27 patients who completed the study subsequently transitioned into the extension study.

Patient characteristics have been described [40], and are shown in Supplementary Table S1. In brief, there were slightly more women (16/29 [55.2%]) than men (13/29 [44.8%]), and the median (range) age was 54.0 (26–72) years. All patients were Japanese, and all had an ECOG PS of 0. Approximately half had previously used HU (15/29 [51.7%]). Twenty-seven patients had a *JAK2* V617F mutation (mean ± SD 72.2% ± 23.3%).

Complete hematologic response

Results for the primary endpoint of durable CHR (lasting at least 12 weeks), for the overall population and according to age and prior HU use, are shown in Table 1.

CHR in the ITT population at 12, 24, 36, and 52 weeks is shown in Fig. 2A. CHRs to ropeginterferon alfa-2b treatment were observed in 1/29 (3.4%) patients at week 12, and increased over time. More than half of the patients (15/29 [51.7%]) had achieved a CHR at week 52, and 8 (27.6%) patients who achieved CHR at week 52 had an experience of achieving CHR by week 24. Eight of 29 (27.6%) patients achieved the primary endpoint (durable CHR without

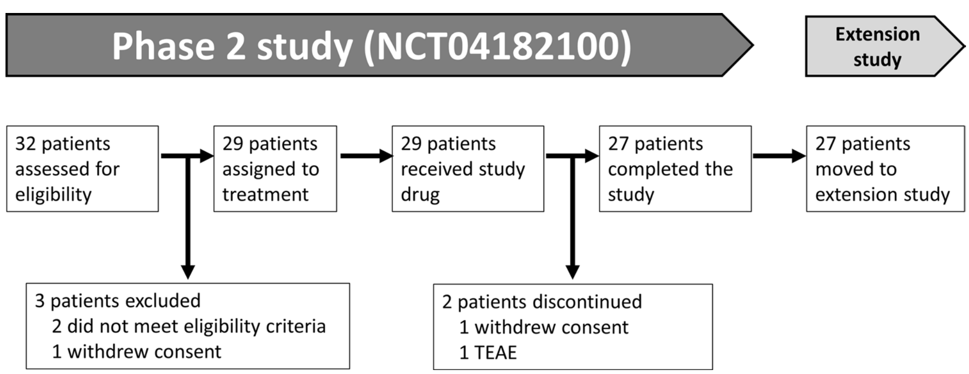

Fig. 1 Patient disposition. *TEAE* treatment-emergent adverse event

Table 1 Durable CHR[a], overall and according to baseline characteristics (intention-to-treat population)

CHR, *n* (%) [95% CI]	All patients $N=29$ (100%)	Age < 60 years $n=22$ (75.9%)	Age ≥ 60 years $n=7$ (24.1%)	With prior HU use $n=15$ (51.7%)	Without prior HU use $n=14$ (48.3%)
Responder	8 (27.6) [12.7, 47.2]	6 (27.3) [10.7, 50.2]	2 (28.6) [3.7, 71.0]	3 (20.0) [4.3, 48.1]	5 (35.7) [12.8, 64.9]
Non-responder	21 (72.4) [52.8, 87.3]	16 (72.7) [49.8, 89.3]	5 (71.4) [29.0, 96.3]	12 (80.0) [51.9, 95.7]	9 (64.3) [35.1, 87.2]

CHR complete hematologic response, *CI* confidence interval, *HU* hydroxyurea

[a]Lasting for at least 12 weeks; assessed via central laboratory

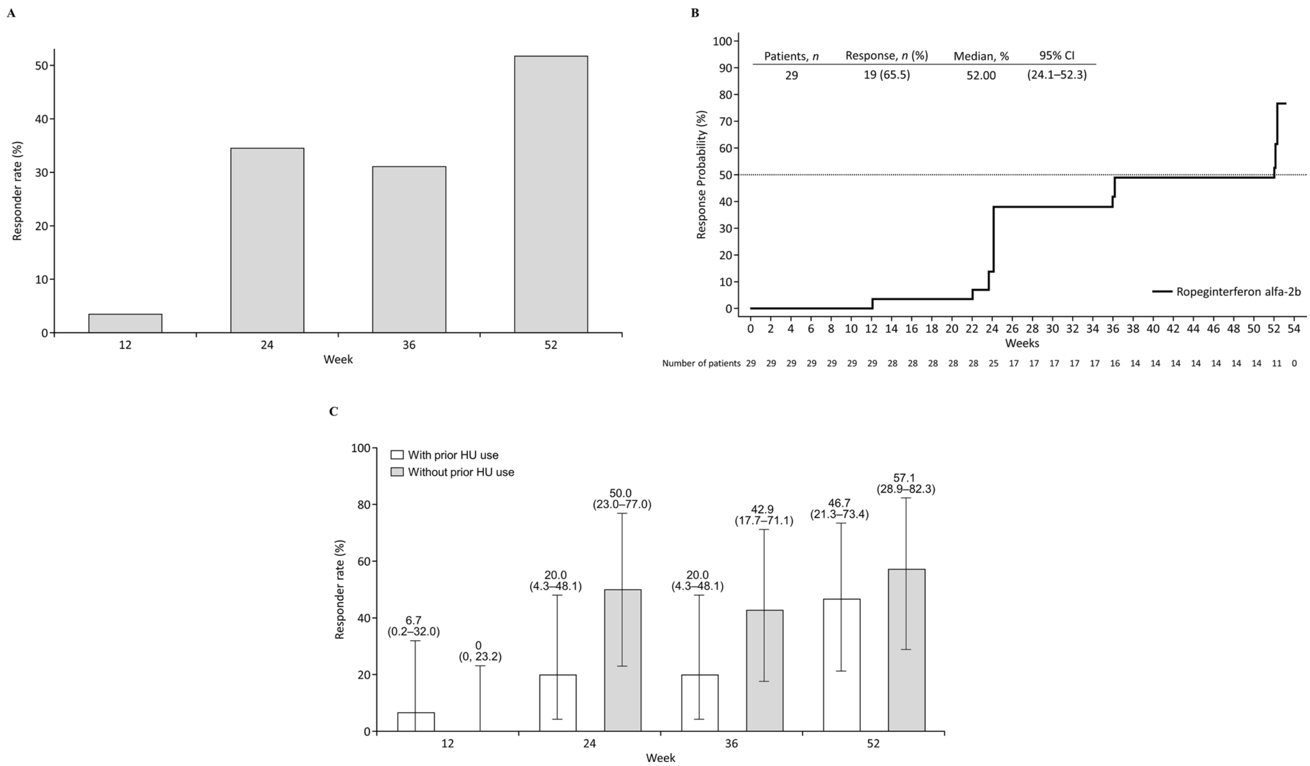

Fig. 2 Complete hematologic response. **A** Proportions of responders over time. **B** Time to response. **C** Difference in response (95% CI) according to prior treatment with HU (intention-to-treat population). *CI* confidence interval, *HU* hydroxyurea

phlebotomy) during the study period (Table 1). As shown in Fig. 2B, the median time to CHR was 52.0 weeks (95% CI: 24.1–52.3). The difference in CHR with or without prior treatment with HU is shown in Fig. 2C. Five out of 14 patients (35.7%) without prior use of HU achieved the primary endpoint vs 3 out of 15 patients (20.0%) with prior use of HU.

Secondary outcomes

Hematocrit, platelet count, and WBC count decreased over 52 weeks of treatment with ropeginterferon alfa-2b (Fig. 3A–C); mean (SD) changes from baseline at the end of treatment were -5.5% (6.7%), -493.6×10^9/L (374.9), and -11.7×10^9/L (8.4), respectively. The mean of all parameters improved to their target value ranges.

The baseline *JAK2* V617F allele burden decreased over 52 weeks of treatment (Table 2), and the rate of molecular response to ropeginterferon alfa-2b increased over the 52-week treatment period (Supplementary Fig. S1). Twenty-seven patients completed the study. Of the 27 patients, 26 patients had a *JAK2* V617F mutation, and changes in the *JAK2* V617F allele burden were analyzed for the 26 patients. The mean (SD) change in allele burden from baseline to end of treatment ($n = 26$) was -19.2% (22.6%). The median (range) allele burden at week 52 ($n = 26$) was 52.5% (6.2%–92.0%). All 13 patients without prior use of HU had decreased *JAK2* V617F allele burden from baseline at week 52, whereas 9 of the 13 patients (69.2%) with prior use of HU had decreased *JAK2* V617F allele burden. Supplementary Fig. S2 illustrates the individual changes in *JAK2* V617F allele burden from baseline to week 52; further details during the observational period are provided in Supplementary Table S2. Twenty-two of 27 patients who completed the study had decreased *JAK2* V617F allele burden at week 52 (twenty-six patients who completed the study had the *JAK2* V617F mutation, as mentioned previously). Of the 12 patients with *JAK2* V617F allele burden $< 50\%$ at week 52, 10 achieved a CHR at week 52, including 6 patients who met the primary endpoint. All four patients whose allele burden was increased at week 52 compared with baseline had been previously treated with HU (Supplementary Fig. S2). Although eight patients were categorized as high-risk (age ≥ 60 years, $n = 7$; history of thrombosis, $n = 1$) at baseline, no relationship was observed between high-risk status and CHR, achievement of the primary endpoint, or changes in *JAK2* V617F allele burden (Supplementary Fig. S2).

The dosing trends with exposure to ropeginterferon alfa-2b over time are shown in Supplementary Fig. S3. Fourteen of 29 patients (48.3%) reached the maximum dose (500 µg) of ropeginterferon alfa-2b by the end of treatment, with more than half achieving this dose within 20 weeks. Of the 15

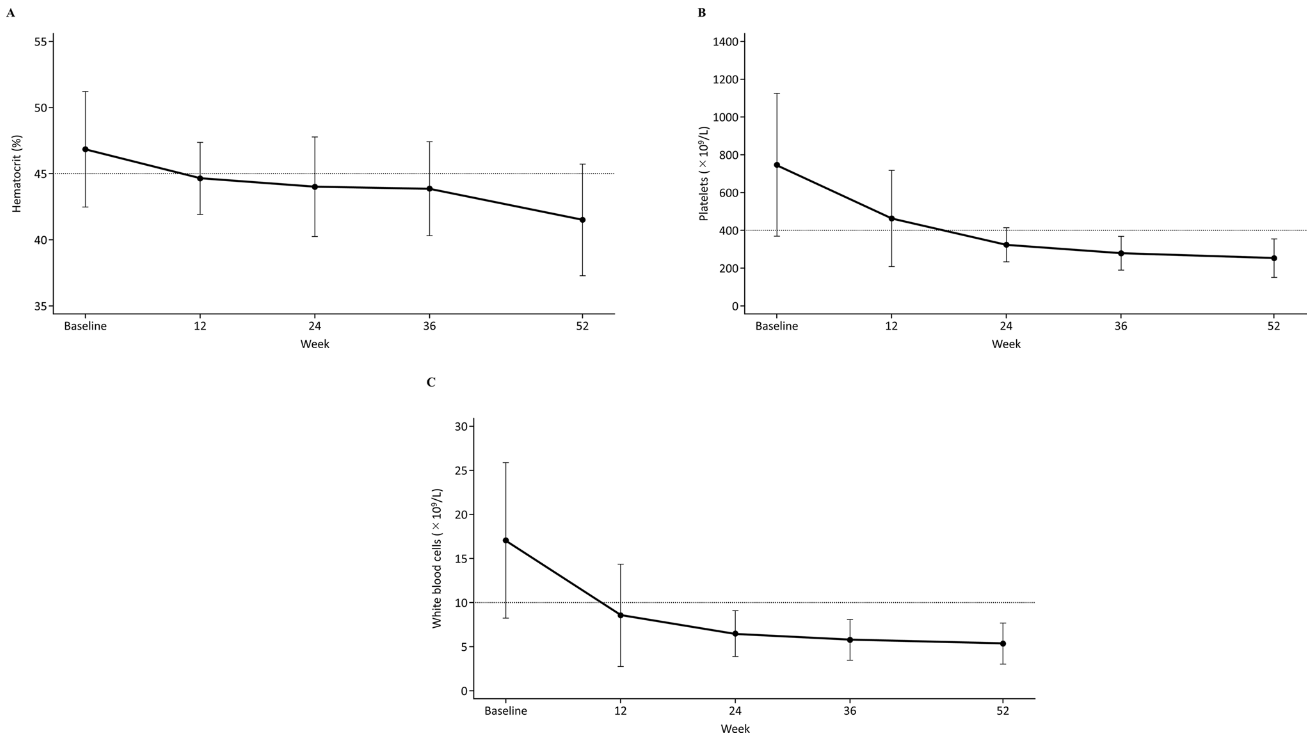

Fig. 3 Laboratory values over time. **A** Hematocrit. **B** Platelet count. **C** White blood cell count (intention-to-treat population). Data are mean (standard deviation)

patients who did not reach the maximum dose, 10 (34.5%) achieved CHRs before reaching the maximum dose (a criterion for stopping dose escalation), 3 (10.3%) had dose reductions due to TEAEs, and 2 (6.9%) withdrew from the study before reaching the maximum dose.

Spleen index and change from baseline by assessment visit for the ITT population are summarized in Supplementary Table S3. No marked change in spleen index was observed in any patient during the 52-week observational period of this study. Additional information regarding spleen size based on longitudinal length is reported in Supplementary Data 2.

Safety outcomes

Safety data are summarized in Table 3 and a full listing is provided in Supplementary Table S4. All patients experienced at least one treatment-related TEAE; however, most were mild or moderate in severity with no TEAEs of grade ≥ 3. One serious TEAE was reported (gastroenteritis) but a causal relationship with the study drug was ruled out and the TEAE did not result in treatment discontinuation. One patient was discontinued due to a TEAE (silent thyroiditis). There were no deaths during the study. The most common TEAEs related to study treatment were alopecia (16/29 [55.2%]), fatigue (8/29 [27.6%]) and influenza-like illness (8/29 [27.6%]). All cases of alopecia were grade 1 (mild), except for one patient who had grade 2 (moderate) alopecia. The median (range) days to onset of alopecia was 177.5 days (43–267). Alopecia had recovered/resolved in four patients by the end of the study and remained "ongoing" at week 52 in 12 of the 16 patients. Individual changes in clinical statuses of patients who developed alopecia are summarized in Supplementary Table S5.

TEAEs of special interest related to study treatment included hypothyroidism in two patients, and silent thyroiditis, anxiety, and retinal hemorrhage occurring in one patient each. Autoimmune thyroid dysfunction was suspected in one of the two patients who developed hypothyroidism and the one patient who developed silent thyroiditis because autoantibodies against thyroglobulin and thyroid peroxidase were detected. A 45-year-old female patient developed grade 1 hypothyroidism at a dose of 500 µg. The dose was not changed, because the event was mild and non-serious. A 57-year-old male patient developed silent thyroiditis (grade 1) at a dose of 300 µg (doses from start to end), and administration of the study drug was subsequently suspended on day 113. Because the TEAE was not recovered/resolved within 8 weeks of the treatment interruption, the administration was discontinued in accordance with the study protocol.

Table 2 *JAK2* V617F allele burden (%) and change from baseline by assessment visit according to ELN criteria in 2009 (intention-to-treat population)

Visit	Value	Change from baseline
Baseline		
N	27	–
Mean (SD)	72.2 (22.8)	–
IQR	53.8–91.4	–
Median (range)	81.8 (23.6–98.3)	–
Week 12		
N	27	27
Mean (SD)	67.3 (21.3)	− 4.9 (10.1)
IQR	50.2–87.8	− 12.3– − 0.2
Median (range)	74.7 (25.4–94.9)	− 5.8 (− 19.4–32.7)
Week 24		
N	27	27
Mean (SD)	64.7 (22.2)	− 7.5 (15.4)
IQR	43.6–84.7	− 14.5– − 2.1
Median (range)	73.2 (24.0–94.6)	− 8.0 (− 36.3–49.9)
Week 36		
N	26	26
Mean (SD)	60.92 (24.28)	− 12.5 (16.3)
IQR	41.4–83.9	− 21.7– − 4.5
Median (range)	67.2 (9.6–92.7)	− 11.6 (− 41.3–44.9)
Week 52		
N	26	26
Mean (SD)	54.2 (28.9)	− 19.2 (22.6)
IQR	31.7–83.7	− 32.6– − 6.4
Median (range)	52.5 (6.2–92.0)	− 11.6 (− 74.8–33.8)

JAK2 Janus kinase 2 gene, *ELN* European LeukemiaNet, *IQR* interquartile range, *SD* standard deviation

PK outcomes

Summary statistics for serum ropeginterferon alfa-2b concentration–time profiles from the current study are presented in Supplementary Tables S6 and S7. Plots of individual observed serum ropeginterferon alfa-2b concentrations over time are shown in Supplementary Figs. S4 and S5. Although the PK data were sparsely sampled, at week 0 (initial administration), serum ropeginterferon alfa-2b concentrations appeared to decline sharply in many patients, suggesting the potential presence of target-mediated drug disposition at the initial doses studied (50 or 100 µg) in the unsaturated condition. However, at week 28, when all patients had their ropeginterferon alfa-2b dose escalated to between 150 and 500 µg, serum ropeginterferon alfa-2b concentrations appeared to decline more slowly and in a mono-exponential manner in all patients after the absorption phase concluded (Supplementary Fig. S5).

Values for serum ropeginterferon alfa-2b $AUC_{0\text{-}t}$ and C_{max} versus dose at week 28 are shown in Fig. 4, along with the corresponding results of the power model regressions. Both $AUC_{0\text{-}t}$ and C_{max} failed to meet the equivalence criteria at week 28 and appeared slightly less than dose-proportional. However, caution is warranted in the interpretation of these results because of the small number of samples analyzed.

Discussion

Ropeginterferon alfa-2b has shown positive clinical results for the treatment of PV. It was approved for this indication in Europe, Taiwan, South Korea, and most recently, in the US [32–35]. However, Japanese clinical data have been lacking. In this phase 2 study, ropeginterferon alfa-2b was shown to be a safe and effective treatment in Japanese patients with PV including patients classified as having a low risk of thrombotic events based on the current guidelines. Eight of 29 patients (27.6%) in our study achieved a durable CHR with ropeginterferon alfa-2b. CHRs were first observed at week 12, with more than half of patients (51.7%) achieving a CHR at week 52.

The data from this study in Japanese patients with PV demonstrated that ropeginterferon alfa-2b had efficacy and safety profiles comparable with previously published studies in Western populations [27–31]. Although the primary endpoint criteria were different, the proportion of patients with durable CHR in this study was broadly comparable with the proportion of European patients who had CHR with normal spleen size (26/122 [21.3%]) in the PROUD-PV study [30]. In PROUD-PV, the rate of CHR in the ropeginterferon alfa-2b group (defined as hematocrit < 45% without phlebotomy [at least 3 months since the last phlebotomy], platelet count ≤ 400 × 10^9/L, WBC count ≤ 10 × 10^9/L and not including the composite spleen criterion) was 53/123 (43%) at 12 months [30]. It is interesting to note that the CHR rate at 12 months in the current study was 51.7%, comparable with that of PROUD-PV. With regards to the changes in the spleen index from baseline, any inferences must be made with caution due to the relatively short observational period (52 weeks) in the current analysis.

A reduction in the *JAK2* V617F allele burden from baseline to 52 weeks after ropeginterferon alfa-2b treatment was also observed in both our study (mean change of − 19.2%) and the PROUD-PV study (mean change of − 11.20%) [30]. These findings are consistent with a prior report investigating treatment with ropeginterferon alfa-2b, in which hematologic response correlated with a lower allele burden [27]. The results of the present study suggest that the reduction of abnormal clones harboring the *JAK2* V617F mutation is associated with the improvement of hematological parameters, as 10 of the 12 patients (including 6 patients who met the primary endpoint) whose allele burden decreased to less than 50% at week

Table 3 Summary of adverse events (safety population)

Summary of events	$N = 29$
Patients with at least one TEAE related to study treatment	29 (100)
Patients with at least one serious TEAE[a]	1 (3.4)
Patients with at least one grade ≥ 3 TEAE related to study treatment	0
Patients with at least one grade 2 TEAE related to study treatment	9 (31.0)
Patients with at least one grade 1 TEAE related to study treatment	20 (69.0)
Patients with at least one TEAE of special interest	9 (31.0)
TEAEs related to study treatment that led to study treatment discontinuation[b]	1 (3.4)
Deaths	0
Events related to study treatment occurring in ≥ 10% of patients by PT	
Alopecia	16 (55.2)
Fatigue	8 (27.6)
Influenza-like illness	8 (27.6)
Alanine aminotransferase increased	6 (20.7)
Beta-2 microglobulin urine increased	6 (20.7)
Aspartate aminotransferase increased	5 (17.2)
Diarrhea	5 (17.2)
Liver function test abnormal	4 (13.8)
Myalgia	4 (13.8)
Pyrexia	4 (13.8)
Anemia	3 (10.3)
Arthralgia	3 (10.3)
Gamma glutamyl transferase increased	3 (10.3)
Injection site reaction	3 (10.3)
Malaise	3 (10.3)
TEAEs of special interest related to study treatment by SOC and PT	
Endocrine disorders	3 (10.3)
Hypothyroidism	2 (6.9)
Silent thyroiditis	1 (3.4)
Psychiatric disorders	2 (6.9)
Anxiety	1 (3.4)
Suicidal ideation	1 (3.4)
Eye disorders	1 (3.4)
Retinal hemorrhage	1 (3.4)

Data are shown as *n* (%) and classified according to the Medical Dictionary for Regulatory Activities, v23.0

[a]Gastroenteritis

[b]Silent thyroiditis

PT preferred term, *SOC* system organ class, *TEAE* treatment-emergent adverse event

52 achieved CHR concurrently. This correlation indicates that interferon alfa-based therapies such as ropeginterferon alfa-2b may have a disease-modifying ability [30, 43]. This is supported by evidence from several studies suggesting that treatment with pegylated interferon preparations can sustainably reduce *JAK2* V617F allele burden over long periods, a disease-modifying activity not elicited by other cytoreductive therapies such as HU [27, 44, 45]. The reduction of *JAK2* V617F in patients with PV is thought to be a critical factor in slowing disease progression, as evidence suggests that an increase in *JAK2* V617F allele burden may promote transformation into secondary myelofibrosis [46].

The safety data from our study did not signal any new concerns in Japanese patients with PV, compared with European patients. All patients in our study experienced at least one TEAE (regardless of causality), compared with 84.6% of ropeginterferon alfa-2b-treated patients in PROUD-PV [30]; however, no Japanese patients experienced a TEAE of grade ≥ 3 in our study, whereas grade 3/4 events were experienced by 33.9% of European patients. However, the incidence of alopecia in our study was higher than that of

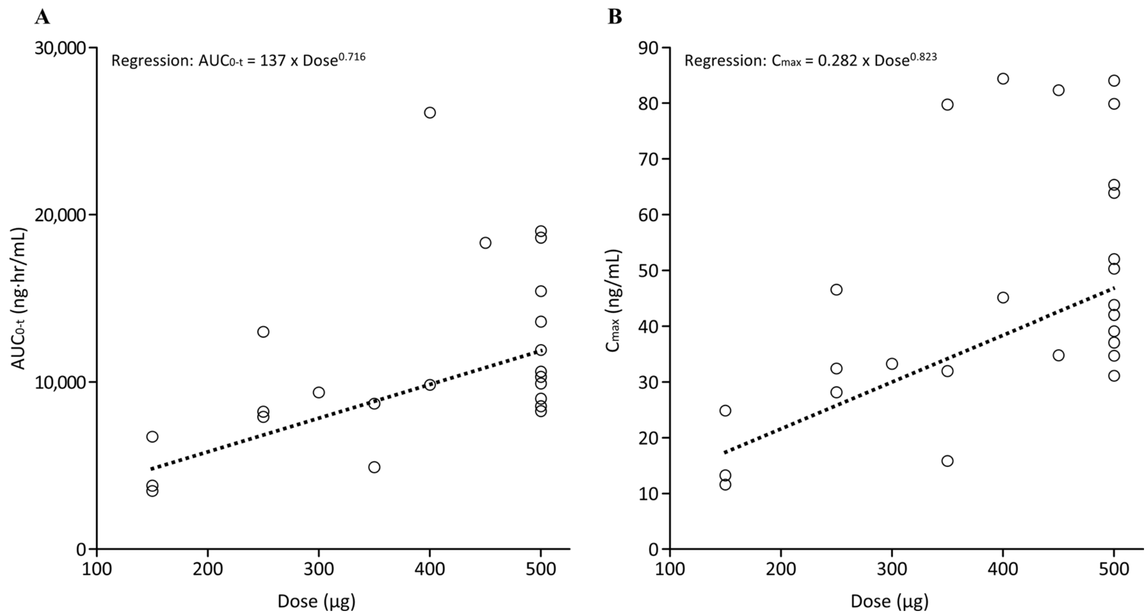

Fig. 4 Serum ropeginterferon alfa-2b $AUC_{0\text{-}t}$ (**A**) and C_{max} (**B**) versus dose at week 28. Only patients with quantifiable serum ropeginterferon alfa-2b data collected 14 (±1) days after dosing were included in the regression analysis. $AUC_{0\text{-}t}$ area under the plasma concentration–time curve from time zero to time t, C_{max}, maximum observed concentration

PROUD-PV (55.2% versus 4.7%) [30]. Alopecia is a known, frequent adverse effect of interferon treatment and has been reported in 19% of patients treated with a combination interferon and ribavirin. It usually manifests as diffuse thinning of the hair, reversible in the majority of cases [47, 48]. Although alopecia was common in our study, all events except for one were graded as mild. Importantly, there were no treatment-related deaths during either study.

Interestingly, a prior multicenter, parallel-group, phase 1 study indicated that the pharmacodynamic and safety findings were similar between healthy Japanese and Caucasian participants, although the ropeginterferon alfa-2b exposure was approximately twofold higher in the Japanese participants [36]. Importantly, the difference in exposure appeared to have little impact on the safety and clinical biomarkers of the study participants. However, it must be noted that the above phase 1 study included healthy volunteers who received a single administration of ropeginterferon alfa-2b, and that the current study is the first to report PK data from Japanese patients with PV who received multiple therapeutic doses. Moreover, this suggests that flexible individual dosing for each patient may allow fine-tuning of the desired efficacy at acceptable tolerability levels. Our findings indicated a target-mediated drug disposition at the initial doses studied, and a change to slower, mono-exponential declines after week 28. Exposure ($AUC_{0\text{-}t}$ and C_{max}) at week 28 failed to meet the equivalence criterion and appeared slightly less than dose-proportional, although the limited number of PK observations confound a definitive data interpretation. Overall, we consider that these data, coupled with the findings from prior studies, suggest no difference in terms of safety and efficacy of ropeginterferon alfa-2b between Japanese and non-Japanese patients with PV.

The improvements in hematologic and molecular responses observed in this study are very important in terms of disease management and control for Japanese patients with PV. Long-term effective treatment options for Japanese patients with PV are urgently needed to prevent the onset of thrombotic complications [7, 8, 49]. Even with the short duration of the current study, the observed decrease in *JAK2* V617F allele burden will have a clinically relevant impact via the decreases in abnormal cytogenetics/tumor burden and the corresponding reduced propensity to progress to myelofibrosis and sAML.

Limitations

Limitations of the study include the small population size and the relatively short observational period of the study given the lifelong duration of the condition. Long-term observation of the effectiveness of treatment with ropeginterferon alfa-2b is required, and there is an ongoing extension study, which will further assess ropeginterferon alfa-2b as a long-term PV treatment option.

Conclusions

This phase 2 study demonstrated ropeginterferon alfa-2b to be a safe and efficacious treatment option in Japanese patients with PV, regardless of patients' estimated risk of thrombosis.

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Declarations

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